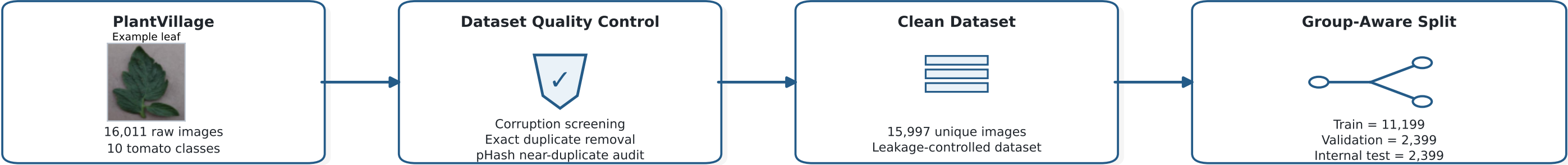
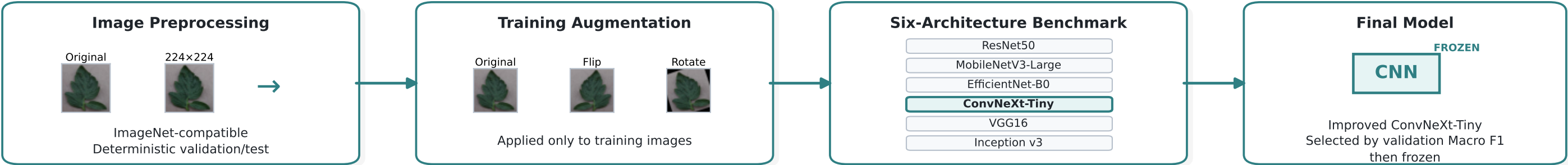


Leakage-Aware Tomato Disease Classification and External Validation Workflow

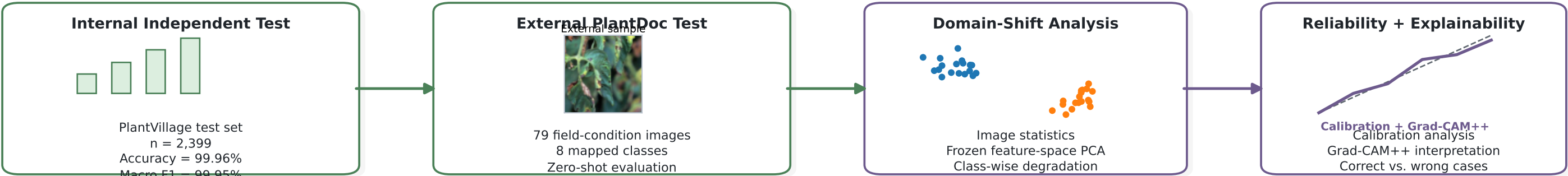
1 Dataset Preparation



2 Model Development



3 Independent Evaluation and Generalization Analysis



Leakage-aware internal benchmarking → frozen external validation → domain-shift and XAI characterization
No PlantDoc fine-tuning, threshold optimization, or external-test-driven model selection